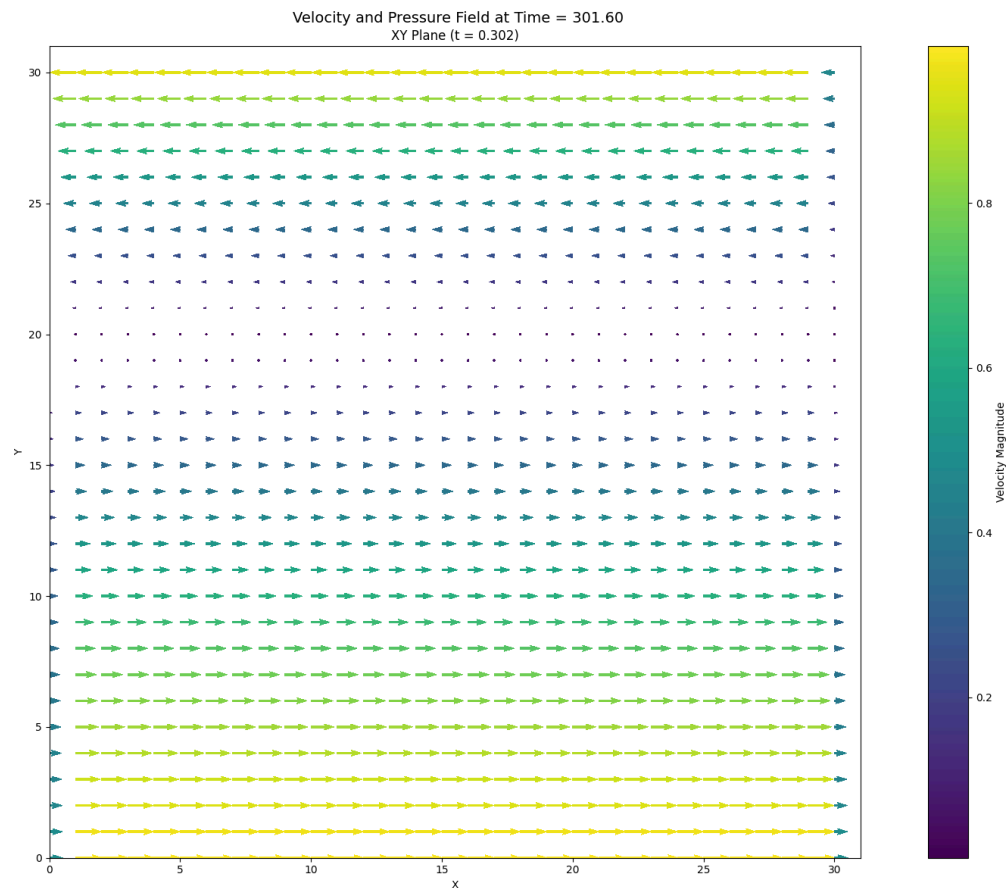


Physics Simulation and STRidge Identification Report

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



1. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	9.20000000e+02	kg m ⁻³
Dynamic viscosity	ν	8.10000000e-02	Pa s
Kinematic viscosity	ν	8.80434783e-05	m ² s ⁻¹
Characteristic velocity	$u_{\max}$	1.00000000e+00	m s ⁻¹
Characteristic length	D	5.00000000e-03	m
Reynolds number	Re	1.97740055e+01	dimensionless
Body force	G	9.81000000e+00	m s ⁻²
Grid points	N_x	3.10000000e+01	points
Unused transverse grid points	N_y	3.10000000e+01	points
Grid spacing	dy	1.61290323e-04	m

Quantity	Symbol	Value	Unit
Time step	dt	1.31646880e-04	s
Numerical threshold	epsilon	1.00000000e-15	-

2. Numerical Discretisation

Quantity	Symbol	Value	Unit
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Parameters listed above are the values used by the selected solver.

3. Data and Pre-processing

Quantity	Value
Loaded field shape (t, y)	(2142, 31)
Regression field shape (t, y)	(542, 27)
Regression samples	14,634
Dictionary terms	7
First saved time index	150
Temporal crop at each boundary	800 layers
x-boundary crop	0 points
y-boundary crop	2 points

4. Governing Equation Verification

Reference equation

$$u_t = 8.80434783e-05 \cdot u_{yy} + 9.81000000e+00 \cdot \text{one}$$

Learned equation

$$u_t = -2.08702206e-10 \cdot u \cdot u_y + 2.68851484e-07 \cdot u + 1.56320160e-08 \cdot u_y + 8.84726794e-05 \cdot u_{yy} + 9.85782285e+00 \cdot \text{one}$$

Term	Reference coefficient	Learned coefficient	Absolute error
$u \cdot u_y$	+0.00000000e+00	-2.08702206e-10	2.08702206e-10
$u \cdot u_{yy}$	+0.00000000e+00	+4.66507141e-12	4.66507141e-12
$u_y \cdot u_{yy}$	+0.00000000e+00	+1.34869663e-13	1.34869663e-13
u	+0.00000000e+00	+2.68851484e-07	2.68851484e-07
u_y	+0.00000000e+00	+1.56320160e-08	1.56320160e-08

Term	Reference coefficient	Learned coefficient	Absolute error
u_yy	+8.80434783e-05	+8.84726794e-05	4.29201163e-07
1	+9.81000000e+00	+9.85782285e+00	4.78228475e-02

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-09
Selected tolerance	1.00000000e-06
L0 penalty	1.00000000e-06
Training RMSE	2.63621763e-09
Validation RMSE	3.34331075e-09

6. Reproducibility Notes

- Input directory:
/Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_2movingwalls/u_velocity_field
- Fields were loaded starting at saved time index 150.
- Derivatives were calculated using second-order edge-aware finite differences via `numpy.gradient`.
- The learned equation is compared term-by-term with the supplied reference coefficients.